

Student number: 1911-142-8
Date: 2026-01-09

LEBOPO B MR
2580 MATSIE STREET
PROTEA NORTH
1818

Dear Student

As requested, a statement is attached.

Yours faithfully

Prof MM Sepota

Registrar



Student number: 1911-142-8
Date: 2026-01-09

This is to certify that

BONGANE LEBOPU

Identity Number :9904255071080
Date of Birth :1999-04-25

passed the university examinations in the undermentioned study units for which credit has been granted in partial completion of the

Bachelor of Science in Informatics

NQF exit level: 7
Minimum credits required: 360

YEAR MONTH	CODE	NAME OF STUDY UNIT	%	NQF LEVEL	CREDITS
2024 FEB	COS1511	Introduction to Programming I	57	5	12
2023 OCT	COS1512	* Introduction to Programming II	87	5	12
2023 OCT	COS1521	Computer Systems: Fundamental Concepts	71	5	12
2023 JUN	FAC1502	Financial Accounting Principles, Concepts and Procedures	50	5	12
2023 OCT	INF1505	Introduction to Business Information Systems	70	5	12
2023 OCT	INF1511	* Visual Programming I	80	5	12
2023 OCT	INF1520	Human-Computer Interaction I	71	5	12
2023 JUN	MNB1501	Business Management IA	64	5	12
2024 OCT	COS2611	* Programming: Data Structures	87	6	12
2024 OCT	COS2614	* Programming: Contemporary Concepts	89	6	12
2025 FEB	COS2626	Computer Networks I	69	6	12
2024 OCT	ICT2621	Structured Systems Analysis and Design	70	6	12
2025 FEB	ICT2622	Object-Oriented Analysis	68	6	12
2025 FEB	INF2603	Databases I	59	6	12
2024 OCT	INF2611	Visual Programming II	71	6	12
2024 JUN	MNB1601	* Business Management IB	85	6	12
2024 JUN	MNG2601	* General Management	86	6	12
2025 FEB	STA1610	Introduction to Statistics	62	6	12
2025 OCT	COS3711	Advanced Programming	57	7	12
2025 OCT	FAC1602	Elementary Financial Accounting and Reporting	72	6	12
2025 OCT	INF3703	* Databases II	82	7	12
2025 JUN	MNE2601	Introduction to Entrepreneurship and Small Business Management	68	6	12
2025 OCT	MNG3701	* Strategic Planning IIIA	91	7	12
2025 OCT	MNG3702	* Strategic Implementation and Control IIIB	83	7	12

* Passed with distinction

Total credits accumulated: 288

Major subject(s): INFORMATION SYSTEMS

This qualification is not completed.

Yours faithfully



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Student number: 1911-142-8
Date: 2026-01-09



Registrar



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Purpose statement of modules passed

This is to certify that the purpose statement of the modules offered comprises the following:

COS1511 - Introduction to Programming I

The purpose of the module is to introduce students to programming and to cover the fundamentals of data and control structures, techniques for problem solving and algorithm design, input and output of data from and to the standard input/output streams. Data types and structures which are covered are floating point, integer, character, string, boolean, one and two dimensional arrays. The C++ decision and iteration structures that are covered include if, while, for, switch and do..while. Furthermore, functions with both reference and value parameters are covered, as well as struts.

COS1512 - Introduction to Programming II

To introduce students to the detailed design and implementation of algorithms as programs, and includes the fundamentals of simple data structures with object-orientation.

COS1521 - Computer Systems: Fundamental Concepts

To introduce students to the computer as a system. This covers hardware concepts such as internal representation of numbers and characters and basic computer architecture, and software concepts such as systems software and applications software. It also includes a brief introduction to databases, and to systems analysis and design.

COS2611 - Programming: Data Structures

To show learners how abstract data types and data structures can be implemented and used in an object-oriented programming language. The module covers recursion, linked lists, dynamic memory allocation, binary trees, and graphs.

COS2614 - Programming: Contemporary Concepts

This module provides qualifying learners with the knowledge, skills and competencies to apply object- oriented programming techniques and strategies in solving real-world problems according to industry- approved processes within South-African and global contexts. The students who complete this module can design and implement object oriented software systems. These abilities prepare qualifying students to be competent programmers that are familiar with structured programming techniques.

COS2626 - Computer Networks I

To introduce the fundamental building blocks that form a modern network, such as protocols, topologies and hardware. This module provides an in-depth coverage of the most important concepts in contemporary networking, such as TCP/IP, Ethernet, wireless transmission, and security. Students will also acquire the skills to build a network from scratch and maintain, upgrade, and troubleshoot an existing network.

COS3711 - Advanced Programming

To increase the depth of student's insight into advanced programming principles and consolidate their competence there-in.



FAC1502 - Financial Accounting Principles, Concepts and Procedures

To equip students with the fundamental concepts, standards and principles of financial accounting and to provide an informed understanding of the collection, processing and recording of relevant financial accounting data of an entity. To enable students to prepare the Statement of profit or loss and other comprehensive income, the Statement of changes in equity and the Statement of financial position. To obtain an understanding of the accounting treatment and disclosure of non-current assets, current assets; non-current liabilities and current liabilities; including the preparation of the financial statements of sole enterprises, non-profit organisations and the recording of transactions from incomplete records.

FAC1602 - Elementary Financial Accounting and Reporting

To equip students with the necessary competencies for the preparation and presentation of financial statements according to the requirements of International Financial Reporting Standards and to gain knowledge and understanding of the theories, relevant in this accounting field, as well as the ability to apply this knowledge to the following subject matter: specific aspects of the Conceptual Framework for the preparation and presentation of financial statements; preparation of financial statements for sole proprietors, partnerships and close corporations; recording of changes in the ownership structure of partnerships on the admittance, retirement or death of partners; preparing statements of cash flows for sole proprietors, partnerships and close corporations; explain how a business entity with branches operates and recording the transactions between head office and branch.

ICT2621 - Structured Systems Analysis and Design

To enable an intermediate level systems analyst to analyse information systems, and design computerized solutions using structured analysis and design techniques.

ICT2622 - Object-Oriented Analysis

To enable an intermediate level systems analyst to analyse information systems according to the object oriented approach using the tools, techniques and methodologies of systems development.

INF1505 - Introduction to Business Information Systems

Qualifying students can apply information technology (IT) concepts in their lives, identify different classes of business information systems, and understand systems development and information systems in business and the wider society. This module provides fundamental introductory knowledge, skills and values which will support further studies and applications in the sector of Information Technology and Computer Sciences and Commerce as part of either the BSc degree or BCom Informatics degree. This module will support further studies and applications in the sector of Computing.

INF1511 - Visual Programming I

Qualifying students as first-time programmers obtain introductory knowledge, skills, and competencies to apply visual programming concepts, techniques and strategies using problem solving, programming logic, as well as the design techniques of an object-oriented, event-driven language, Python. This module forms part of a B-degree and supports further studies and applications in the sector of Computing, in the fields of Computer Science, Information Systems or Multimedia. The qualifying student can programme computers to solve problems in business and society within African, South-African, and global contexts. Students require daily online connectivity and access and programming ability.

INF1520 - Human-Computer Interaction I

Students who complete this module successfully will have a fundamental-level overview of the principles and concepts of human-computer interaction (HCI). Students become cognisant with trends in the development of usercentred computer applications and in the types of interfaces and interaction styles. They gain an understanding of various attributes of the intended users that may influence computer use, such as their cognitive, perceptual, cultural and social characteristics. Students will be able to describe and compare different usability evaluation methods. They will be qualified to serve as novice members of a design team of interactive technological systems.



INF2603 - Databases I

Qualifying students can understand, design and manage database management systems. This module serves as a fundamental building block in equipping students with the knowledge and competencies to understand and use databases. This module provides fundamental and required knowledge, skills and values which will support further studies in the field of Database Management design and implementation systems on NQF level 7 as part of either Bachelor of Science degree in Computing or Informatics or the Bachelor of Commerce degree in Business Informatics or the Bachelor of Commerce degree with Informatics as major. These competencies contribute to the development of competitive information technology practitioners who have strong technical skills in designing, implementing and managing database systems. The module contributes to the development of the computing field in Southern Africa, Africa, or globally. Students are required to have computers and Internet access.

INF2611 - Visual Programming II

Qualifying students apply knowledge, skills and competencies in order to develop menu-driven, multiple layout- and database -applications in the visual programming paradigm, through the use of Python. The qualifying student is able to programme computers to solve problems in business and society within African, South-African and global contexts. Students require daily online connectivity and access and programming ability.

INF3703 - Databases II

The purpose of the module is to equip students with adequate knowledge and competencies to design, implement, and manage centralized and distributed database systems. Students registered for this module are introduced to the database development processes, database administration roles and responsibilities, stages of database design, database transactions, database performance tuning, distributed database concepts, database security, distributed databases within the cloud environment, decision support systems, and strategies for interacting with databases through the web. This module provides fundamental and required knowledge, which will support further studies and applications in the sector of Computer Science and Information systems, in the field of Database Management Systems as part of either Bachelor of Science degree in Computing or Informatics or the Bachelor of Commerce degree with Informatics as major. These competencies therefore contribute to the development of competitive Information Technology practitioners who have strong technical skills of designing, implementing and managing single and distributed database systems.

MNB1501 - Business Management IA

The purpose of this module is to establish a foundation for the study of Business management; and to develop the learner's understanding of the concept of strategic management and of the specific management tasks (planning, organising, leadership, control). This will enable qualifying learners to develop basic business plans that reflect an entrepreneurial approach to maximising the use of resources. These business plans are of the type that could be used for seeking funding from financing institutions.

MNB1601 - Business Management IB

The purpose of this module is to provide insight and an understanding of the individual functional areas of business management which ultimately form a synergistic whole that directs the business towards its goals. These functional areas are human resources, marketing, finance, operations and purchasing and supply management. This module will provide the foundation for learner's further study in the aforementioned functional areas of a business

MNE2601 - Introduction to Entrepreneurship and Small Business Management

The purpose of the module is to empower students with the graduate attributes needed analyse the uniqueness of the relationship between entrepreneurship and the small business in emerging economies. Students will also be able to contribute to job creation and economic wealth and growth through sound managerial, governance, entrepreneurial practices and innovation. In addition, students will investigate alternative routes to start a business including the uniqueness of a family business and will also gain knowledge on how to develop their business idea into an opportunity that can provide a successful product or service in emerging economies.



MNG2601 - General Management

Successful students will demonstrate critical managerial competencies and skills to implement the general principles of planning, organising, leading and control, relevant to a changing environment, to ensure a sustainable and responsible organisation that makes an ethically sound and positive contribution to the broader society.

MNG3701 - Strategic Planning IIIA

Students who complete this module will be able to conceptualise and explain relevant components of strategic planning as part of the strategic management process. They will be able to evaluate the strategic direction of an organisation and recommend suitable business-level strategies based on a thorough environmental analysis.

MNG3702 - Strategic Implementation and Control IIIB

Students who complete this module can analyse and critically evaluate the choices made in implementing and controlling business-level strategy. They will be able to think strategically in practical situations when making recommendations on responsible strategy implementation and strategic risk, control and governance measures. In addition, students will be able to implement and control strategies in a responsible manner, based on sustainability, responsibility towards all stakeholders, and ethical decision making.

STA1610 - Introduction to Statistics

To ensure that students are introduced to the most important basic statistical concepts. After completion students should have an informed understanding of different visual descriptions of data, including graphical and tabular techniques; measures of central location, dispersion and association. They should be able to use probability as a tool to create discrete and continuous probability distributions, used extensively in statistical inference; determine confidence intervals and perform hypothesis testing involving sample means and proportions; apply different forms of Chi-square testing; understand simple linear regression and correlation.

